Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

David Afonso Valente *dvalente@bottlecapai.com*
Bottlecap AI

Abstract

Base models are usually chosen by public benchmark scores, which vendors optimise for, which can leak into training data, and which grade a model before the fine-tuning it will receive. We study a label-free alternative: rank the candidates by how well they predict fresh, unlabelled text from the target domain, in bits per byte. How the text is scored decides whether this works. Scored on whole documents, each read from its own beginning, the ranking places all fifteen base models within one rank of where they finish once fine-tuned on a news-writing task (Spearman 0.99), against 0.97 for HellaSwag and 0.59–0.71 for GSM8K and MMLU-Pro. Scored the common way, on fixed-length blocks cut from concatenated text, it misranks several models (Spearman 0.71), and some of the strongest models look weak. Adapting every model briefly to the domain repairs the block reading; on whole documents the reading needs no adaptation. The adapted ranking agrees with HellaSwag on general text, and on arXiv mathematics it agrees more closely with GSM8K and MMLU-Pro than with HellaSwag. It is stable across adaptation seeds and hardware. Code and artifacts: https://github.com/DavidAfonsoValente/entropy-bench.

1 Introduction

Suppose an engineer has to pick a base model for a product. They have plenty of text from the product's domain and a shortlist of a dozen open checkpoints, and the model they pick will be fine-tuned and shipped. The most reliable way to choose would be to fine-tune every candidate and compare the results, but that needs a task definition, labelled data and a training run for each model. In practice the choice is made from benchmark scores instead: MMLU-Pro (Wang et al., 2024), multiple-choice questions across academic subjects; HellaSwag (Zellers et al., 2019), which asks for the most natural ending of a passage and is scored by likelihood; and GSM8K (Cobbe et al., 2021), grade-school maths word problems. In this paper we show that, on a news-writing task, the fine-tuning ranking can be anticipated before any task exists, from unlabelled text alone. We rank the candidates by how well they predict held-out text from the domain, scoring each document whole, from its own beginning. The resulting ranking matches the ranking after fine-tuning almost exactly. Of the benchmarks, only HellaSwag comes close.

Benchmark scores can mislead for two reasons. The first is that benchmarks are public. They were written for other purposes and released before the models they now evaluate, vendors optimise for the ones they report, and their questions can end up in pre-training data. A high score may come from a stronger model or from a model that has seen more of the test, and because pre-training data is rarely disclosed, a reader cannot tell which. Benchmarks can also disagree sharply with how well a model handles ordinary text. Llama-3.2-1B and Qwen-2.5-1.5B are 2.1 points apart on HellaSwag and their loss on news text after adaptation differs by only 3.4%, but they are 55.3 points apart on GSM8K (accuracy, in percentage points).

The second reason is that models are rarely used as they ship. Teams fine-tune a base model on their own data, so what matters is how good it becomes, and a benchmark score taken before that can misjudge it. Loss on the domain's text can misjudge it too, depending on how the text is scored. Gemma-4-12B is an example. Among the fifteen fine-tuned models it ranks ninth on GSM8K, eighth on MMLU-Pro and thirteenth on loss scored the common way, on fixed-length blocks cut from concatenated text. It ranks second on loss scored on whole documents, and it produces the best of the fifteen fine-tuned systems (Table 2).

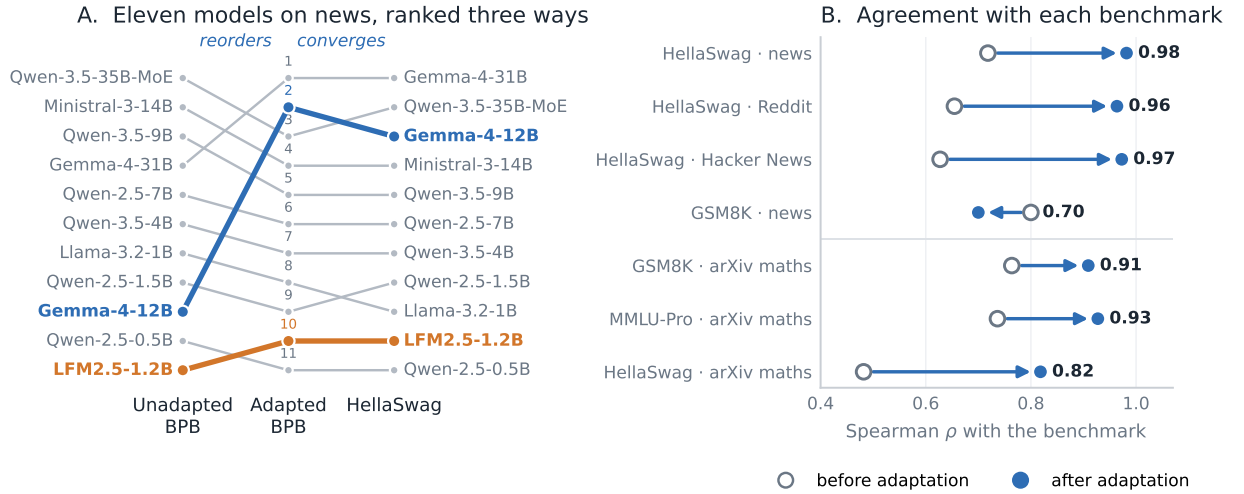


Figure 1: Adaptation reorders the block reading, and the text determines what the ranking follows. **A**: the eleven models evaluated on all four corpora, ranked by news bits per byte (BPB; lower is better) scored on 512-token blocks before adaptation, after adaptation (rank shown above each point), and by HellaSwag. Many lines cross between the first two columns, as Gemma-4-12B moves from ninth to second and LFM2.5-1.2B leaves last place. Between the last two columns the lines are almost parallel. **B**: rank agreement (Spearman ρ; 1 means lower BPB goes with a higher score) between BPB and a benchmark, before adaptation (hollow) and after (filled). On general text adaptation raises agreement with HellaSwag and lowers it with GSM8K. On arXiv mathematics it raises agreement with all three benchmarks, and the ranking ends closer to GSM8K and MMLU-Pro than to HellaSwag. The full matrix is Table 5 in Appendix A.

Loss is a natural measure for this choice. There is no universal ground truth for which base model is better: a benchmark tells us which model is better at that benchmark, and different benchmarks disagree. A base model acquires its capabilities by minimising cross-entropy on text, and adaptation minimises the same loss on the domain. Held-out loss is therefore the objective each candidate was last trained on, measured on text it was not trained on. Cross-entropy is also a strictly proper scoring rule (Gneiting & Raftery, 2007). If p is the distribution that generated the domain's text and q is a model, the model's expected loss is

$$H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q), \tag{1}$$

where $H(p)$, the entropy of the text, is the same for every model, and $\mathrm{KL}(p \,\|\, q)$ measures how far the model's predictions are from the text's distribution (Cover & Thomas, 2006). The difference between two models' expected losses is therefore the difference in their KL divergence from the text, and held-out loss estimates it. Equation 1 says which model fits the domain's text best; whether that predicts usefulness on a task is an empirical question, and Section 5.1 tests it on one task.

Three design choices keep the comparison fair. We count the loss per byte instead of per token, because each model has its own tokenizer but they all read the same bytes; we call the result *bits per byte* (BPB). We compute it on fresh, decontaminated text that none of the candidates has trained on. And we score each document whole, from the start the model was trained to see, on exactly the same bytes for every model. We began with a different third choice: adapting every candidate in the same way to the domain before scoring fixed-length blocks. That reading ranks the fine-tuned models just as well, and we report both, because comparing them shows what adaptation was correcting.

We evaluate seventeen base models from eight families, from 0.5B to 35B parameters. Eleven are evaluated on four corpora (news, Reddit, Hacker News and arXiv mathematics), and fifteen are also fine-tuned on a news task, writing an article's opening paragraph from its body. Table 8 lists which model enters which analysis. Our main findings are:

- **BPB on the domain's text anticipates fine-tuning on the task.** When the models are ranked by BPB on whole documents before any fine-tuning, every one of the fifteen ends up within one rank of its

position after fine-tuning. Adapting the models first and scoring blocks does as well. HellaSwag comes close, and GSM8K and MMLU-Pro are at chance (Figure 3, Section 5.1).

- **How the text is cut can reorder the models.** Scored on blocks cut mid-document, some strong models look weak, Gemma-4-12B most of all. Adaptation repairs this, and so does scoring each document from its own start, with no adaptation (Section 5.2).
- **The choice of text changes what the ranking follows.** On general text the adapted ranking agrees closely with HellaSwag; on arXiv mathematics it agrees more closely with GSM8K and MMLU-Pro than with HellaSwag (Figure 1B, Section 5.3).
- **The ranking is stable.** No size-matched pair changes order across adaptation seeds, and re-runs on other hardware reproduce the numbers (Section 5.4).

All numbers in the paper can be recomputed from the released artifacts with `make check`:

`https://github.com/DavidAfonsoValente/entropy-bench`

The paper and its artifacts are also attached to each release (`https://github.com/DavidAfonsoValente/entropy-bench/releases/latest`).

2 Related Work

Loss, compression and capability. Language modelling was grounded in the entropy of text from the start (Shannon, 1951). Token-level perplexity depends on each model's vocabulary (Mielke, 2019; Ahia et al., 2023); normalising by UTF-8 bytes restores a common unit (Gao et al., 2021; Magnusson et al., 2023), and loss-based metrics carry more signal and less noise than accuracy across hundreds of models (Heineman et al., 2025). Language modelling is equivalent to compression (Delétang et al., 2024), and compression of fresh text correlates linearly with benchmark scores across about thirty models (Huang et al., 2024). UncheatableEval (UncheatableEval contributors, 2024) maintains a leaderboard of compression on recently published text for the same reason we use fresh text. These works, like our main reading, measure models as they ship. We check the ranking against fine-tuned systems rather than against benchmarks alone, and show that how the text is cut into evaluation windows can change it. The metric is not new; the contribution is the protocol around it and what it shows.

Choosing a model to fine-tune. Lin et al. predict which LLM will fine-tune best by fitting a rectified scaling law to small fine-tuning runs on the target task (Lin et al., 2024); this needs task data, where our measurement needs only unlabelled text. Transferability estimators such as LEEP and LogME (Nguyen et al., 2020; You et al., 2021) score a pre-trained model from its features on labelled target data. Train-before-test (Zhang et al., 2025a) gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark agreement; the idea of preparing every candidate identically before comparing them is theirs, and we do it without labels, on fresh text, in a unit that compares tokenizers. Observational scaling laws relate benchmark performance across model families to a few capability measures (Ruan et al., 2024), and loss predicts downstream error within controlled families (Gadre et al., 2025). Thrush et al. use the varying correlation between per-domain perplexity and benchmark scores to select pre-training data (Thrush et al., 2024).

Adapting before measuring. Dynamic evaluation updates a model on the recent history of the sequence it is being evaluated on (Krause et al., 2018); domain-adaptive pre-training improves fit to a target distribution (Gururangan et al., 2020). Our adapted reading adapts once, on a training split, before any test text is scored.

Contamination. Models memorise duplicated pre-training text (Carlini et al., 2021; 2023), so deduplication and contamination analysis matter (Lee et al., 2022; Xu et al., 2024). Reference-free membership signals (Shi et al., 2024; Zhang et al., 2025b) are weak for large language models (Duan et al., 2024), which is why our main protection is the date of the text and the screens serve only for conservative removal. Fresh benchmarks such as LiveBench (White et al., 2025) apply the same idea to labelled questions.

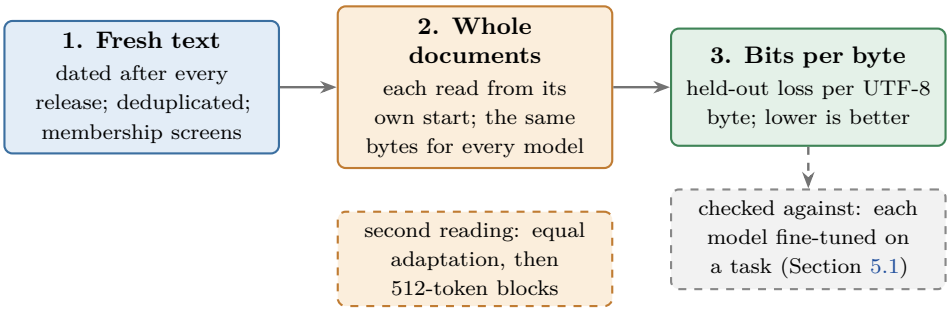


Figure 2: The method. The corpus is fixed and screened first; every candidate then reads the held-out documents whole and is scored in bits per byte. The second reading adapts every candidate in the same way before scoring blocks. The fine-tuning experiment of Section 5.1 is not part of the method. We use it only to check the method's predictions.

3 Method

The method has three phases (Figure 2). The corpus is collected and screened before any model is run on it, then every candidate reads the held-out documents whole, and its loss is counted in bits per byte. We also report a second reading, in which every candidate is first adapted in the same way and then scored on fixed-length blocks. We scored both readings on news; the adapted block reading is the one we have on all four corpora.

Phase 1: a decontaminated corpus. The evaluation text must be text that no candidate has seen, and we use three safeguards for this. *Time:* Reddit and Hacker News were scraped in August 2026, after every model in the cohort was released, and the arXiv titles and abstracts were posted between 9 June and 20 August, after most of the releases. *Hygiene:* we remove exact duplicates, text that fails the Gopher quality heuristics (Rae et al., 2021) and MinHash/LSH near-duplicates (Broder, 1997) at Jaccard 0.60. The 80/10/10 split shares no identifier or normalised text across splits. *Membership screens:* the news snapshot (Google News, 8 June 2026, 119,054 articles) predates some of the releases, so we also test every model for having seen it. Min-K%++ (Zhang et al., 2025b) flags passages whose least likely tokens are unusually likely for the model, and a context-sensitivity test flags passages whose loss drops more than expected when the model is shown other text from the same corpus. We remove every passage flagged for any model, 80 in total, so all candidates are measured on the same corpus (Appendix B). The corpora, their manifests and record fingerprints are released.[1]

Phase 2: whole documents. Each test document is read on its own, starting from its tokenizer's beginning-of-document token, if it adds one, with up to 2,048 tokens of context. A document longer than that is read in overlapping windows with a stride of 1,024, and each token is scored once. The beginning-of-document token is never scored. Every model reads the same first 6.0 MB of test documents, so the text is identical byte for byte. The implementation is released.[2] The per-model scores are released with the analysis that turns them into the numbers below.[3]

Phase 3: bits per byte. Let $\mathcal{L}$ be the total cross-entropy, in nats, over the scored tokens of the test text, and $|B|$ the UTF-8 byte length of the text those tokens cover:

$$\mathrm{BPB} = \frac{\mathcal{L}}{|B| \ln 2}. \tag{2}$$

A model with a larger vocabulary predicts fewer, longer tokens; dividing by bytes rather than tokens removes the penalty per-token perplexity would give it.

[1] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/docs/DATASET.md
[2] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/lm_adapt_bench/evaluate.py
[3] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results/doc_protocol_cells

The adapted block reading. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, using LoRA (Hu et al., 2022) on all linear layers, and then scored on non-overlapping 512-token test blocks cut from the packed test stream, each read with no earlier context. Beginning-of-document tokens inside a block are used as context but never scored. Because blocks are cut in tokens, each model covers a slightly different amount of text, 5–10% apart (Appendix B). The same block reading without adaptation is the *unadapted block* reading. For adaptation, each candidate can get its own hyperparameter search with Optuna (Akiba et al., 2019). A Tree-structured Parzen Estimator (TPE; Bergstra et al., 2011) proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving (Jamieson & Talwalkar, 2016) prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never on the test split. Every model gets the same number of trials and no time limit, because with a wall-clock limit small models complete more trials than large ones and end up better tuned (Appendix B). The best configuration is then trained until validation BPB stops improving. The released pipeline implements this procedure.[4] Every adapted number in this paper uses a single shared configuration instead, for compute reasons (Section 4). Adapted models can also read whole documents, and we report that too.

4 Experimental Setup

Models. Eleven base checkpoints between 0.5B and 35B parameters are evaluated on all four corpora (Table 9). Most are dense Transformers; one is a sparse mixture-of-experts, and LiquidAI's LFM2.5-1.2B is a hybrid of convolution and attention blocks. For the news corpus we add six more models, chosen before they were run to fill gaps in model size and to add publishers, for 17 in total (Table 8). They come from the Llama, Qwen, Gemma, Ministral, Mistral-NeMo, Falcon, SmolLM and LFM families (Dubey et al., 2024; Meta AI, 2024; Yang et al., 2024; Qwen Team, 2026; Gemma Team, 2026; Liu et al., 2026; Mistral AI, 2024; Technology Innovation Institute, 2024; Ben Allal et al., 2025; Liquid AI, 2026). Seven of the seventeen are Qwen models.

Corpora. We use four corpora: news, Reddit, Hacker News and arXiv mathematics. Table 4 gives their sizes and how each was screened.

Adaptation runs. Because of limited compute, all runs reported here use one shared configuration instead of the per-model search: LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch size 32 and 250 update steps, a common LoRA setting. The runs are a fixed budget, not trained to convergence. This is 4.1M training tokens per candidate and costs 3.7 GPU-minutes on one A100 at 0.5B parameters, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B; at 512 tokens per sequence every model's adaptation fits on one 64 GB GPU. The budget is matched in tokens, so models with larger vocabularies see a few percent more bytes of text. Its results are stable across seeds and hardware (Section 5.4).

Benchmarks. For comparison, we score every model on MMLU-Pro (a stratified 1,000-question subset, so one question is 0.1 points), HellaSwag (all 10,042 validation items) and GSM8K (all 1,319 test items) with *lm-evaluation-harness* 0.4.12 (Gao et al., 2024), on one machine with one software build, following the protocol in Appendix E. We use scores we measured ourselves, not the values reported on model cards, and we publish the harness logs.

The fine-tuning task. We fine-tune fifteen of the seventeen models (Table 8), each starting from its base weights, to write an article's opening paragraph, its lede, from the rest of the article. Gemma-4-31B and Qwen-3.5-35B-MoE are left out because the fine-tune's 1,280-token sequences and generation pass do not fit them on one GPU. All fifteen use the same LoRA recipe as the adaptation and are scored by ROUGE-L, an F-measure based on the longest common subsequence with the reference (Lin, 2004), on 500 held-out articles that are not part of the BPB test set. The pairs to compare and the statistic were fixed in code before the fine-tuning jobs ran (Appendix C). We chose this task because every article provides its own reference, the lede written by the journalist, so no annotation is needed, and writing a good lede requires reading the whole article and writing in the style of the domain.

[4] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/lm_adapt_bench

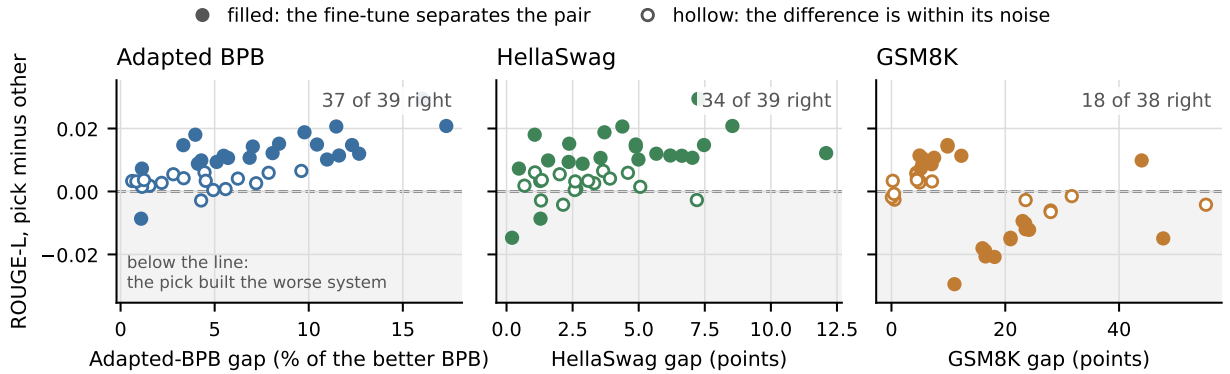


Figure 3: How well does each ranking predict the fine-tuned results? Each point is a pair of fine-tuned models whose parameter counts are within a factor of two (39 pairs; 38 for GSM8K, where two models tie). For each pair, x is how strongly the ranking prefers one model (the gap in that measure) and y is how much better that model's fine-tuned system scores in ROUGE-L. Points above the line are pairs the ranking orders correctly. Filled markers are *resolved* pairs, whose ROUGE-L difference has a 95% bootstrap interval over the 500 articles that excludes zero. BPB on whole documents gets 37 of 39 pairs right, including 21 of the 22 resolved pairs, and adapted block BPB, plotted here, orders the pairs identically; HellaSwag gets 34; GSM8K gets 18 of 38, and all eleven of its pairs with gaps between 15 and 25 points fall below the line.

Statistics. We compare rankings mainly by *pairwise accuracy*, the fraction of model pairs that a ranking orders in the same way as a reference. Pairwise accuracy counts only *size-matched* pairs, two models whose parameter counts differ by less than a factor of two; when one model is more than twice the size of the other, the larger one usually wins, and a ranking would get the pair right by following size. Because pairs that share a model are not independent, 95% intervals come from a cluster bootstrap over models (20,000 draws). Intervals for a single fine-tuned pair resample its 500 evaluation articles. With fifteen models the intervals are wide, and we say so where it matters.

5 Results

5.1 BPB on the domain's text anticipates fine-tuning on the task

A team will fine-tune whichever model it chooses, so the practical question is whether the cheap measurement can anticipate how that fine-tuning will turn out. Measured with no task, no labels and no training, BPB on whole documents orders 37 of the 39 size-matched pairs (0.949, 95% interval $[0.817, 1.000]$) in the same way as the fine-tuned systems. HellaSwag orders 34 (0.872, $[0.640, 1.000]$), BPB on blocks without adaptation 30 (0.769), and GSM8K and MMLU-Pro are at chance with 0.474 and 0.513 (Figure 3, Table 1). The intervals of BPB and HellaSwag overlap, so the data do not separate these two. On the 22 pairs that the fine-tuning results themselves resolve, the document reading gets 21 right. Adapted block BPB, the reading we started with, ranks the fifteen models in exactly the same order, so every ordering statistic in this section is the same for both. Adapting the models and then reading whole documents orders 38 of the 39 (0.974, $[0.862, 1.000]$) and all 22 resolved pairs; Section 5.2 gives the one pair it changes. One pair out of 39, with intervals that do not separate the two readings, does not pay for training every candidate, so the unadapted document reading is our main one. Seven of the 39 pairs differ by less than 2% in adapted block BPB, inside the range we treat as a tie (Section 5.4); the band comes from the seed noise of adaptation, so it does not apply to the document reading, which has none. Outside that band adapted block BPB orders 31 of 32, and its miss there (Ministral-3-14B against Qwen-3.5-9B) is a pair the fine-tuned systems do not resolve. Its one resolved miss, Mistral-Nemo-12B against Gemma-4-12B, is a 1.1% gap inside the band. Over all fifteen models the document ranking agrees with the final order at Spearman 0.99: every model is within one rank of its final position, and eleven are exactly in place. For comparison, unadapted block BPB reaches 0.71, GSM8K 0.59, MMLU-Pro 0.71 and HellaSwag 0.97. BPB also beats ranking by model size, by +0.231 in pairwise accuracy $[+0.059, +0.452]$.

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB, documents	**0.974**	[0.862, 1.000]	1.000	0.897
Unadapted BPB, documents	0.949	[0.817, 1.000]	0.955	0.923
Adapted BPB, blocks	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Unadapted BPB, blocks	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 1: Which ranking predicts the fine-tuned systems? Fifteen models, each fine-tuned with the same recipe on the lede task and scored by ROUGE-L on 500 held-out articles. Each row gives the pairwise accuracy of one ranking over all size-matched pairs, with a bootstrap interval over models, and its accuracy on the 22 pairs whose fine-tuned difference is resolved by a bootstrap over articles. The last column scores the same rankings against systems trained with recipe B. GSM8K ties on one pair and is scored on 38.

	Before fine-tuning			After	BPB, documents
Model	BPB, blocks	GSM8K	MMLU-Pro	**fine-tuning**	**predicted**
Gemma-4-12B	13	9	8	**1**	2
Llama-3.2-1B	10	15	15	**11**	11
LFM2.5-1.2B	15	12	13	**14**	14

Table 2: Three models the benchmarks misjudge. Rank among the fifteen fine-tuned models (1 is best). Gemma-4-12B is mid-table on both benchmarks and near the bottom on unadapted block BPB, then builds the best system of all; Llama-3.2-1B is last on both benchmarks and finishes eleventh. BPB on whole documents, with no adaptation, places every model within a single rank of where it finishes: all fifteen, and eleven exactly. Adapted block BPB gives the same ranks.

A natural objection to the adapted reading is that the fine-tune reuses the adaptation's own recipe, so a model that adapts quickly under it could win both. The objection does not apply to the document reading, which involves no training. With a deliberately different fine-tuning recipe, recipe B (LoRA rank 64 and α=128, three times the learning rate, half the batch size, 400 steps and a new seed), the order of the fine-tuned systems hardly changes (Spearman 0.982). Document BPB and adapted block BPB are still the best predictors at 0.923, and they order correctly all 20 pairs that recipe B separates.

Since the adaptation and the fine-tuning both use news, one might suspect that news is just predicting news. An exploratory check, run after the main result, suggests otherwise. For the nine models that were both fine-tuned and adapted on all four corpora (10 size-matched pairs), BPB adapted on Reddit orders 9 of the 10 pairs like the news fine-tune, the same as news BPB (Spearman 0.983 each); Hacker News orders 7 and arXiv mathematics 6, and the unadapted readings 5–6 (Appendix C). Ten pairs support no interval, so this is a direction, not a result.

Table 2 shows three models that the benchmarks misjudge. Gemma-4-12B, ninth on GSM8K and eighth on MMLU-Pro among the fifteen, produces the best fine-tuned system; in the pair fixed in advance it beats Ministral-3-14B by +0.018 ROUGE-L [+0.011, +0.025]. The 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B did not separate their fine-tuned systems: that pair was inconclusive. Every model improved on its own base weights and beat the score of copying the article's first body sentence (ROUGE-L 0.125).

5.2 How the text is cut reorders the models

The block reading without adaptation ranks the models differently, and adaptation changes that order (Figure 1A, Table 3). For the eleven models, the block rankings before and after adaptation have a Spearman correlation of only 0.655 on news, 0.609 on Reddit and 0.682 on Hacker News, and one model moves seven places. The changes are uneven. On news, adaptation lowers block loss by 3.6–5.4% for every Qwen, Llama and Ministral model, but by 17.0% for Gemma-4-31B, 24.5% for Gemma-4-12B and 36.4% for LFM2.5-1.2B.

Model	Params (B)	News BPB (rank)		Moves	Gain	Benchmark score, % (rank)		
		unadapted	adapted			HellaSwag	GSM8K	MMLU-Pro
Gemma-4-31B	31.4	0.752 (4)	0.624 (1)	↑**3**	**17.0%**	87.9 (1)	85.0 (3)	57.1 (2)
Gemma-4-12B	12.0	0.866 (9)	0.653 (2)	↑**7**	**24.5%**	84.0 (3)	64.8 (7)	44.8 (7)
Qwen-3.5-35B-MoE*	34.7	0.707 (1)	0.678 (3)	↓2	4.1%	85.6 (2)	86.6 (1)	61.4 (1)
Ministral-3-14B	14.0	0.718 (2)	0.679 (4)	↓2	5.4%	82.9 (4)	80.8 (6)	52.8 (4)
Qwen-3.5-9B	9.0	0.740 (3)	0.708 (5)	↓2	4.4%	81.6 (5)	85.8 (2)	55.6 (3)
Qwen-2.5-7B	7.7	0.758 (5)	0.717 (6)	↓1	5.4%	80.3 (6)	81.3 (4)	47.2 (6)
Qwen-3.5-4B	4.2	0.782 (6)	0.749 (7)	↓1	4.2%	77.2 (7)	81.0 (5)	52.4 (5)
Llama-3.2-1B	1.2	0.811 (7)	0.771 (8)	↓1	4.9%	65.8 (9)	6.4 (11)	11.2 (11)
Qwen-2.5-1.5B	1.6	0.833 (8)	0.797 (9)	↓1	4.3%	68.0 (8)	61.6 (8)	25.6 (8)
LFM2.5-1.2B	1.2	1.340 (11)	0.852 (10)	↑1	**36.4%**	60.9 (10)	54.1 (9)	16.4 (9)
Qwen-2.5-0.5B	0.5	0.935 (10)	0.901 (11)	↓1	3.6%	51.8 (11)	34.3 (10)	16.0 (10)

Table 3: The eleven models ranked by every measure. Sorted by adapted news BPB (lower is better). *Moves* is the number of places a model gains or loses with adaptation, and *Gain* is the reduction in loss. Three models gain far more than the rest (the two Gemmas and LFM2.5-1.2B), and they are the only ones that move up. After adaptation the order is almost the same as HellaSwag's, while GSM8K and MMLU-Pro differ: Llama-3.2-1B, eighth by BPB and ninth on HellaSwag, is last on both. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters.

These three are the only models that move up, and every other model drops one or two places. Gemma-4-12B is ninth of eleven before adaptation and second after; before adaptation it ranks below two models with fewer than 2B parameters. LFM2.5-1.2B is last before adaptation at 1.34 bits per byte, behind a 0.5B Qwen at 0.93, and ahead of it after adaptation.

Figure 4 shows where the block reading goes wrong. Before adaptation, Gemma-4-12B starts a block at 2.83 bits per byte, while Qwen-2.5-1.5B, eight times smaller, starts at 2.04. Gemma stays behind for most of the block and is ahead by the end, 0.72 against 0.77. The early positions have the highest loss, so they contribute most of the difference between the two block averages. Blocks are cut from a packed stream, so most start in the middle of a document, without the beginning-of-document token that models such as Gemma are trained to see first.

Two tests, with the decision rule fixed before they ran, separate the cut from the model. First, scoring only positions 64, 128 or 256 onwards of each block removes the first tokens of the block but not the mid-document start; on the 10 size-matched fine-tuned pairs whose models we scored this way, it leaves unadapted block BPB at 6 correct, unchanged, against 9 for adapted block BPB. Second, we read every test document whole, as in Section 3. Without any adaptation this ranks the fifteen fine-tuned models in exactly the order adapted block BPB does (Spearman 1.00), and Gemma-4-12B moves from thirteenth of fifteen to second. Most of what adaptation did to the block reading was teaching the models to read text that starts mid-document. We did not separate the beginning-of-document token, the document boundary and the longer context; together, they amount to reading the text as the model was trained to see it.

On whole documents, adaptation still lowers every model's loss, by 2.0–7.9%, but it hardly changes the order (Spearman 0.993 between the two document readings). Gemma-4-12B again gains the most, and its 7.9% moves it past Mistral-Nemo-12B into first place, which is the one change and the one resolved pair it corrects. The control models reproduce the published block readings on the new host: unadapted to within 0.04%, and adapted to within 0.7%, inside the largest gap seed noise can open (Section 5.4).

5.3 The choice of text changes what the ranking follows

This section and the next use the adapted block reading, the one we ran on all four corpora. On news it orders the fifteen fine-tuned models exactly as the document reading does (Spearman 1.00); whether the document reading follows the same corpus dependence is untested.

Of the three benchmarks, HellaSwag is the closest to language modelling. It asks which ending of a passage is most natural, and the harness scores each ending by its likelihood. Before adaptation, BPB on news, Reddit and Hacker News has a Spearman correlation of 0.63–0.72 with HellaSwag across the eleven models; after adaptation it is 0.96–0.98. On news, the adapted ranking orders 53 of the 55 model pairs the same way as

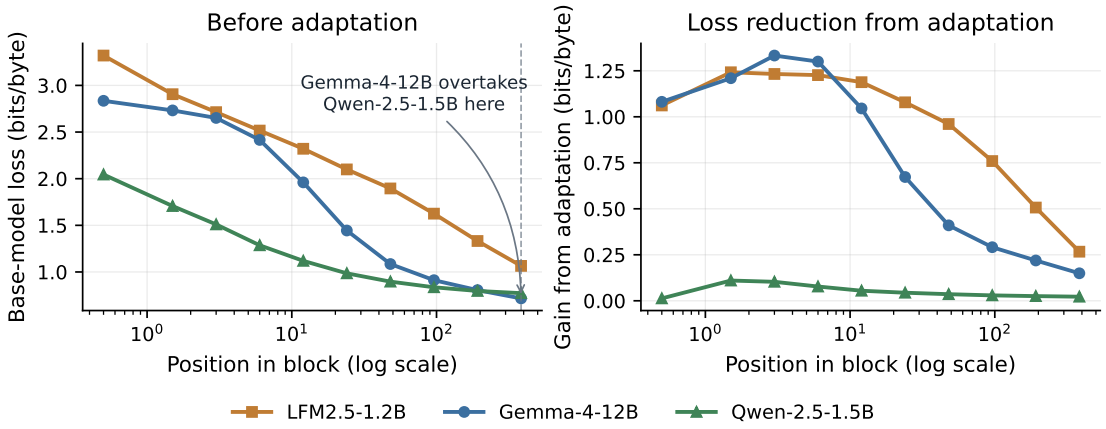


Figure 4: Before adaptation, the start of each block costs some models far more than others. *Left:* loss before adaptation by position within a 512-token evaluation block, for Gemma-4-12B, Qwen-2.5-1.5B and LFM2.5-1.2B. Gemma-4-12B starts a block at 2.83 bits per byte, against 2.04 for Qwen-2.5-1.5B, and ends it at 0.72 against 0.77. *Right:* the loss reduction from adaptation at each position. It is largest at the start of the block but stays positive later. This panel uses adapters from the earlier per-model search (Appendix B), so its gains differ from the shared-recipe gains in Table 3; only the pattern across positions matters here.

HellaSwag, counting all pairs and not only size-matched ones (Table 5 in Appendix A). Since both HellaSwag and BPB are based on likelihood, some agreement is expected; adaptation is what makes it close. On general text, adaptation also lowers the correlation with GSM8K on all three corpora and with MMLU-Pro on two.

When the models are adapted on arXiv mathematics instead, agreement rises with all three benchmarks: with GSM8K from 0.76 to 0.91, with MMLU-Pro from 0.74 to 0.93, and with HellaSwag from 0.48 to 0.82. The rise is therefore not specific to the reasoning benchmarks. What changes is which benchmark the ranking sits closest to: arXiv mathematics is the only corpus on which the adapted ranking agrees more with GSM8K and MMLU-Pro than with HellaSwag, and ranking by mathematics BPB selects GSM8K's top model. This shift is suggestive rather than established. A difference in differences that tests it directly is positive in 93.7% of model resamples (one-sided $p \approx 0.06$) and in 80.9% when whole publisher families are resampled ($p \approx 0.19$), and it depends on the two Gemma models (Appendix B).

5.4 BPB rankings are stable

A ranking is only useful if it is repeatable. The document reading involves no randomness: the adapted jobs re-read every unadapted model and reproduced its document BPB exactly. The adapted reading depends on the adaptation's seed, so we ran three seed replicates of all 17 news adaptations; the seed sets the adapter's initialisation, dropout and batch order, and the data splits and test set stay fixed. None of the 40 size-matched pairs changes order across seeds. The median standard deviation of adapted BPB within a model is 0.0001, against 0.074 between models. The largest gap seed noise could open between two runs is 1.24% of the median BPB ($2\sqrt{2}$ standard deviations of the noisiest model, Gemma-4-12B; the median model, under 0.04%), so we treat a gap under 2% as a tie. Re-running one model, Ministral-3-14B, on different hardware weeks later reproduced unadapted BPB exactly and adapted BPB to within 0.0003, and repeating a run with a different software stack reproduced it to within 0.01%. Rankings also carry over across general text: adapted rankings agree at $\rho = 0.991$ between news and Reddit, 0.955 between news and Hacker News and 0.964 between Reddit and Hacker News (Table 7 and Figure 5, Appendix B). The unadapted ranking is just as consistent across these corpora (0.982, 0.964 and 0.973). One might expect that adaptation buys stability, and it does not: it produces a different consistent ranking, the one that anticipates the fine-tuned order in Section 5.1.

BPB has a structural advantage over generative benchmarks here. Floating-point addition is not associative, so results on a GPU can change with batch size and kernel choice (He & Thinking Machines Lab, 2025). A generative benchmark decodes greedily, so a tiny change in two nearly tied logits can flip a token, and the model then continues from its own changed output. BPB is computed with teacher forcing: every prediction

is conditioned on the true text, so a small perturbation changes one log-probability slightly and does not propagate, and it is averaged over about 1.3 million token predictions. Generative scores are also sensitive to decoding settings that BPB does not use: dropping a single decoding setting shipped with Gemma-4-12B changed its GSM8K score by 9.86 points and its MMLU-Pro score by 14.30 (Appendix E). On hardware alone the effect is small: the same weights move by -0.60 to $+0.69$ GSM8K points between an A100 and an H100, within the benchmark's sampling error (each of its 1,319 questions is worth 0.076 points).

6 Limitations

We test one downstream task, lede writing in news, with two fine-tuning recipes; other task types and domains may behave differently. The document reading was run on news only, for the fifteen fine-tuned models, and on one host; the results across corpora, seeds and hardware are for the adapted block reading. The cohort has fifteen fine-tuned models, seven of them Qwen, so the intervals are wide and BPB and HellaSwag are not separated. The two largest models were not fine-tuned, so the top of the ranking is not checked against a fine-tuned system. All adapted runs use one shared adaptation recipe rather than a per-model search. The shift towards GSM8K and MMLU-Pro under mathematics adaptation rests mainly on two Gemma models. Membership screens cannot prove that a proprietary training set excludes a text; the dates of the Reddit, Hacker News and arXiv corpora are the stronger guarantee.

7 Conclusion

Public benchmarks are a convenient way to choose a base model, but they are public, so they can be optimised for and can leak into training data, and they grade a model before the adaptation it will receive. We ranked candidates instead by their loss, in bits per byte, on fresh, decontaminated text from the target domain, with each document read whole from its own start. It needs no labels, no task definition and no training, only a forward pass over a few megabytes of text.

On a news-writing task, this ranking placed all fifteen fine-tuned systems within one rank of where they finished (Spearman 0.99), against 0.59–0.71 for GSM8K and MMLU-Pro and 0.97 for HellaSwag. How the text is scored matters as much as which text. Cut into fixed-length blocks, the same text misranks several models, and adapting every model first repairs that reading. Gemma-4-12B is thirteenth of fifteen on blocks, second on whole documents, and produces the best of the fifteen fine-tuned systems. The choice of text changes which benchmark the adapted ranking resembles: HellaSwag on general text (Spearman 0.96–0.98), GSM8K and MMLU-Pro on mathematics. The ranking is repeatable across seeds and hardware. A new corpus can be collected for every domain and every evaluation, so the text it is measured on stays unseen.

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A The Full Benchmark Alignment

Table 5 reports all twenty-four (corpus × reading × benchmark) cells over the full 55 pairs, and Table 6 gives, for each reading, how many benchmark points its top-ranked model gives up against each benchmark's own top model. Adapting on general text raises HellaSwag agreement on every corpus and lowers GSM8K agreement on every corpus and MMLU-Pro agreement on two of the three. Adapting on mathematics raises all three. Rank correlations hide how close these are in pair counts: across its eight rows the GSM8K column spans only 41 to 49 of 55 pairs.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K%++ / context / perturbation union
Reddit, 6–10 Aug 2026[†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv `math.*`, 9 Jun–20 Aug 2026[‡]	3,207	3,207	–	15 MB	Post-release scrape + GSM8K shingle screen

Table 4: Corpora. All four receive identical exact deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in contamination control. The membership screens ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort's release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set. Reddit comments are concatenated within a subreddit, and Hacker News items in sequence, into documents of about 600 words, which is why they have fewer documents than raw records. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

		Rank correlation (ρ) with			Pairs ordered the same way, of 55		
Corpus	**Reading**	GSM8K	MMLU-Pro	HellaSwag	GSM8K	MMLU-Pro	HellaSwag
News	zero-shot	0.800	0.845	0.718	45	45	44
	adapted	0.700	0.773	0.982	42	44	53
Reddit	zero-shot	0.745	0.782	0.655	43	43	42
	adapted	0.655	0.745	0.964	41	43	52
Hacker News	zero-shot	0.800	0.809	0.627	44	44	43
	adapted	0.745	0.855	0.973	44	46	53
Math (arXiv)	zero-shot	0.764	0.736	0.482	43	43	40
	adapted	0.909	0.927	0.818	49	49	46

Table 5: Agreement of each BPB reading with each benchmark. Rank correlation of BPB with each benchmark over the eleven models, before (zero-shot) and after adaptation, on every corpus; ρ is signed so that 1 means lower BPB goes with a higher score. On the three general corpora adaptation lifts HellaSwag from 0.63–0.72 to 0.96–0.98 while GSM8K falls; on arXiv mathematics it lifts all three, and GSM8K and MMLU-Pro end above HellaSwag. The last three columns count, of all 55 pairs of the eleven (not only size-matched ones), those that BPB orders the same way as each benchmark: adapted news BPB agrees with HellaSwag on 53.

B Robustness

Consistency across domains. All eleven models ran on three independently collected general-text corpora under the same recipe (Tables 4 and 9). Adaptation changes the ranking: unadapted and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places. The adapted ranking then holds across corpora. News and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news and Hacker News give 0.955 and Reddit and Hacker News 0.964, and the values agree as well as the ranks (Figure 5). The unadapted ranking is as consistent (0.982, 0.964 and 0.973). Adaptation produces a different consistent ranking, and Section 5.1 shows it is the one closer to the fine-tuned order.

Why the search gives every model the same number of trials. The search of Phase 2 uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6}–10^{-3}, log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5}–10^{-1}, log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and promotes the best third at each rung; and the objective is BPB on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (`results/legacy_sweep/*/trials.csv`), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. Apart from the right panel of Figure 4, none of those adapters produces a number in the paper.

Corpus and reading	Picks	Accuracy given up (points)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 6: The cost of the top-1 pick on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that model is below each benchmark's own top model. Only three distinct picks occur in eight rows. Adapted BPB on every general-text corpus picks Gemma-4-31B, the HellaSwag leader, at 1.6 GSM8K and 4.3 MMLU-Pro points. Adapted mathematics BPB picks Qwen-3.5-35B-MoE, the GSM8K and MMLU-Pro leader, which unadapted BPB on news, Hacker News and arXiv also picks.

Corpora compared	**Rank agreement** (Spearman)		**Value agreement** (Pearson)	
	unadapted	adapted	unadapted	adapted
News–Reddit	0.982	0.991	0.996	0.992
News–Hacker News	0.964	0.955	0.994	0.977
News–arXiv maths	0.909	0.755	0.964	0.839
Reddit–Hacker News	0.973	0.964	0.998	0.990
Reddit–arXiv maths	0.918	0.736	0.971	0.882
Hacker News–arXiv maths	0.973	0.836	0.982	0.930

Table 7: Agreement between corpora. Agreement between the eleven models' BPB on two corpora. The three general-text corpora rank the models almost identically before and after adaptation. Mathematics agrees less with the general corpora after adaptation (0.74–0.84 adapted) than before (0.91–0.97 unadapted).

Membership screens. The news corpus passes three reference-free membership screens per model: Min-K%++ (normalised likelihood of the lowest-probability K=20% target tokens), context sensitivity (target loss with and without sampled same-corpus prefixes) and, for flagged passages only, preference for the original passage over two word-swap perturbations. The union over models and screens removed 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the original is expected from fluency alone, so we draw no memorisation conclusion from the third test (`results/contamination/`).

Context length. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens, both models measured improve with context and the larger improves faster, so a protocol that scores only short contexts under-rates models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, the constant the bound below uses. The document reading of Section 3 scores up to 2,048 tokens of context.

Token-aligned blocks: a small bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text: 5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at 0.08× the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the worst ratio of bound to gap is 0.61, for Qwen2.5-3B over Falcon3-7B; the bound overturns no rank. The document reading covers identical bytes for every model and removes the question.

Seed variation is negligible. The published adaptation cells did not fix the `torch` random seed, which sets the LoRA initialisation, dropout and batch order (the data splits are seeded separately). A check fixed before it ran re-ran all 17 news models with three explicit seeds (51 cells, everything else pinned). The median within-model standard deviation of adapted BPB is 0.0001 against a between-model standard deviation of

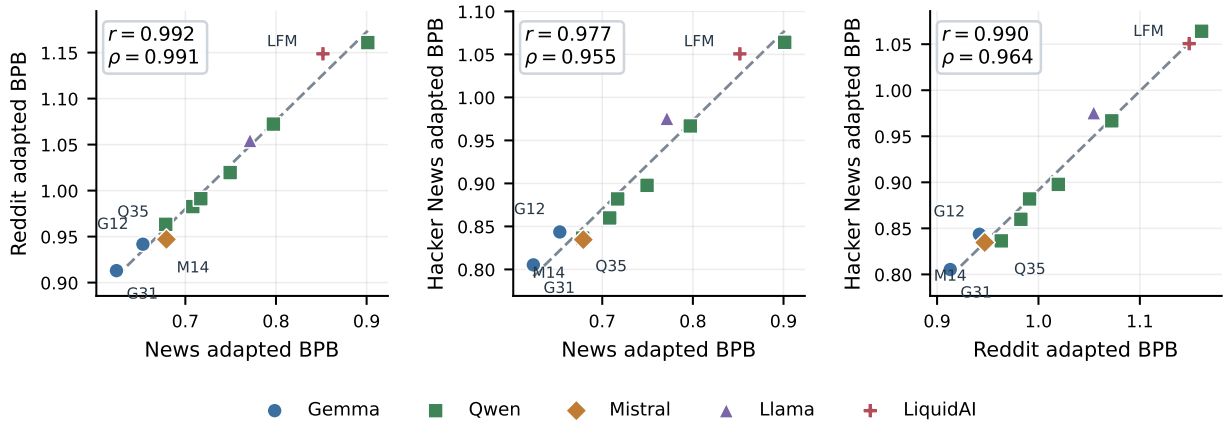


Figure 5: Adapted BPB across domains. Each point is one model, its adapted BPB on two corpora. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between the values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B, Q35 Qwen-3.5-35B-MoE, M14 Ministral-3-14B and LFM LFM2.5-1.2B.

0.074, and none of the 40 pairs of models within 2× in size reverses its order across seeds. Gemma-4-12B is the one model where seed noise is visible (standard deviation 0.003, five times the next largest).

The mathematics shift. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on arXiv mathematics raises GSM8K agreement to 0.909 and recovers GSM8K's top model, while the three general-text corpora all leave GSM8K at or below its unadapted agreement. The shift predicts a *joint* movement, so we test it with a difference in differences: (GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus, which is +0.242 pooled. Resampling models gives a strictly positive difference in 93.7% of draws (one-sided $p \approx 0.06$). Resampling whole publisher families, which the shared training data of a family calls for, gives 80.9% ($p \approx 0.19$), with ties counted as non-positive; not one of its 20,000 draws is negative. A leave-one-family-out jackknife shows the dependence: dropping the two Gemma models takes the pooled difference from +0.242 to +0.056. More families, rather than more models, would settle it.

Reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced unadapted BPB exactly and adapted BPB to within 0.0003. Because the six added models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through that stack; the cohorts are pooled only because it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170).

C The Downstream Fine-Tune

Section 5.1 reports the result; this section describes how it was built. It ran in two stages, each fixed in code before its job returned: first the two pairs on which the rankings disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is hand-picked. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU for this fine-tune, because its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation (Section 4), and are excluded by that rule, which costs one size-matched pair of forty. The fine-tuned scores, pairs and rankings are released.[5]

The task. Journalistic convention makes a news article's first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated

[5] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results/downstream.json

to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The BPB test blocks live in the *test* split, so the fine-tuning task is built from the validation split instead. Evaluation documents are hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four candidate articles were dropped by that second check, before the 500 were sampled, and none by the first.

Equal preparation. Every model gets the adaptation recipe unchanged (LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch 32, 250 steps), starting from its base weights, with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model's realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint's own generation configuration otherwise untouched, following Appendix E. The prompt is plain text: the article body followed by an instruction to write its opening paragraph. When the prompt is too long, the body is cut, never the instruction after it, because cutting the assembled prompt would delete the instruction from the longest articles, at a different article count for each tokenizer. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01; an interval excluding zero resolves the pair for whichever ranking predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort the fine-tuned ROUGE-L is the outcome, and every ranking is scored by pairwise accuracy within 2$\times$ with a model-level bootstrap of 20,000 draws (Table 1). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between +0.041 and +0.153.

A second recipe. Recipe B changes every setting the adaptation and the fine-tune shared (LoRA rank 64, α=128, learning rate 3×10^{-4}, effective batch 16, 400 steps, a new seed) on the same task, data, decoding and metric, and was fixed before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted block BPB, and document BPB with the same order, remain the top ranking at 0.923 $[0.750, 1.000]$, ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here ($+0.179$ $[-0.036, +0.441]$), so that margin is recipe-specific (Table 1).

What copying scores. A news body restates its own lede, so reference overlap is partly available by copying. The article's first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying (inspection confirms they largely do copy), while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result: on the ten size-matched pairs of the nine fine-tuned models that have all four corpora, adapted BPB orders the news fine-tune with pairwise accuracy 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted block readings reach 0.500–0.600. Ten pairs support no interval, so this is a direction, not a result.

D The Cohorts

Table 8 is the 17-model cohort: every model, its parameter count, its adapted news BPB and the analyses it enters. Table 9 gives the original eleven across all three general corpora and both readings.

Model	Params (B)	Adapted BPB	Cohort
Gemma-4-31B	31.4	0.624	original
Mistral-Nemo-12B	12.3	0.646	**added**
Gemma-4-12B	12.0	0.653	original
Qwen-2.5-14B	14.8	0.675	**added**
Qwen-3.5-35B-MoE*	34.7	0.678	original
Ministral-3-14B	14.0	0.679	original
Qwen-3.5-9B	9.0	0.708	original
Qwen-2.5-7B	7.7	0.717	original
Falcon3-10B	10.4	0.728	**added**
Qwen-3.5-4B	4.2	0.749	original
Qwen-2.5-3B	3.1	0.756	**added**
Falcon3-7B	7.5	0.758	**added**
Llama-3.2-1B	1.2	0.771	original
SmolLM2-1.7B	1.7	0.788	**added**
Qwen-2.5-1.5B	1.6	0.797	original
LFM2.5-1.2B	1.2	0.852	original
Qwen-2.5-0.5B	0.5	0.901	original

Table 8: The seventeen models. Ordered by adapted news BPB. "original" marks the eleven models evaluated on all four corpora (Figure 1, Tables 3, 5 and 9); "added" marks the six models added for the news corpus on size-adjacency and publisher-spread criteria fixed before they ran. All seventeen enter the seed replicates; all except Gemma-4-31B and Qwen-3.5-35B-MoE enter the fine-tuning experiment. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

	News		Reddit		Hacker News	
Model	unadapted	adapted	unadapted	adapted	unadapted	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LFM2.5-1.2B	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 9: BPB of the eleven models on the general-text corpora. Lower is better. Beginning-of-document tokens are context only and never scored. Rows are ordered by adapted news BPB. The leaderboard is maintained at https://github.com/DavidAfonsoValente/entropy-bench/blob/main/LEADERBOARD.md.

Three checkpoints are excluded. The two OLMo-2 models fell under a rule fixed in advance for infrastructure faults: their BPB cells exist, but the generation engine shared by the benchmark and fine-tuning evaluations would not initialise for them, so they have no benchmark or fine-tuned scores. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window.

E The Benchmark Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test

examples. All bf16, greedy, with each tokenizer's own beginning-of-sequence token. The raw harness logs for all three benchmarks are released.[6]

One host. Generative scores do not pool across hosts: identical released weights move −0.60 to +0.69 GSM8K points when only the GPU changes from A100 to H100. The benchmark columns were therefore scored again in full on one host (A100-SXM-64GB) with one software build: seventeen models × three benchmarks = fifty-one cells, with the design and the exclusion rule fixed in advance. The task definitions match the earlier scoring exactly (identical task hashes for all seventeen models on GSM8K and HellaSwag; MMLU-Pro ships none), and every overlapping cell agrees with it to within 1.3 points, most to within 0.3.

Decoding settings. The newer harness honours each checkpoint's shipped `generation_config.json`, and the earlier one did not. Replacing every checkpoint's generation settings with one identical policy also discarded `suppress_tokens`, which is set for Gemma-4-12B; that omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points, while HellaSwag, scored by likelihood, was unchanged. Keeping each checkpoint's own settings for all seventeen models reproduces the earlier scoring for Gemma-4-12B to within +1.21 points on GSM8K, +0.10 on MMLU-Pro and −0.27 on HellaSwag. The GSM8K cell is the one outside the 1.0-point tolerance fixed in advance, within the allowance of one such cell.

[6] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results